TEAM ELEVATE TECH | A3-PS003-TC068

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FLOWPULSE-FL

COMMUNICATION-AWARE FEDERATED TRAFFIC-FLOW FORECASTING

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WHAT IS FLOWPULSE-FL?

- FlowPulse-FL is a federated traffic-flow forecasting simulator.
- It represents traffic intersections as simulated clients. Each client trains locally using its own traffic data and shares model updates—not raw records—with a central server.
- The server combines selected client updates using FedAvg to improve a shared forecasting model. FlowPulse-FL also studies whether selecting only some clients per round can reduce communication cost.

PRESENTATION ROADMAP

Problem & Motivation

FL Background

FlowPulse-FL approach

Architecture, data flow, and verified three-seed evaluation

From real-world traffic challenges to a federated, communication-aware forecasting system — here is how the story unfolds across this deck.

THE PROBLEM: DISTRIBUTED TRAFFIC DATA



- Traffic-flow data is generated at separate locations and may remain distributed across local systems.
- Raw data collection at one central location can create privacy and data-governance concerns.
- •Sending every local dataset or every client update in every training round can increase communication cost.
- A shared forecasting model is still useful, but it should be trained without pooling raw client data.
- FlowPulse-FL addresses this by training locally and aggregating only selected model updates.



WHY FEDERATED TRAFFIC FORECASTING?

- FlowPulse-FL models traffic intersections as federated clients.
- Each client trains a local traffic-flow forecasting model using its own data. Raw records stay local; only model updates are shared with the server.
- The server combines updates from selected clients using FedAvg. This lets the shared model improve without collecting all client data in one place.
- FlowPulse-FL compares full participation, random selection, and FlowPulse client selection to study the accuracy–communication trade-off.



HOW FLOWPULSE-FL WORKS

- The server starts each round with a shared global forecasting model.
- A client-selection strategy chooses participating traffic clients: full participation, random selection, or FlowPulse selection.
- Selected clients train locally and send model updates to the server.
- The server combines selected updates using FedAvg and begins the next round with the updated global model.

SYSTEM ARCHITECTURE SERVER–CLIENT WORKFLOW



CLIENT NODES

- Four simulated traffic intersections act as federated clients.
- Each client receives the current global model, trains locally on its own traffic-data partition, and sends a model update only when selected for that round.
- Raw traffic records remain local.

FEDERATED SERVER

- The server selects clients according to the experiment strategy: full, random, or FlowPulse.
- It combines the selected client updates using FedAvg, produces the next global model, and starts the next communication round.

TECHNICAL CONTRIBUTIONS

Federated Learning

- Traffic intersections are represented as simulated federated clients. Each client trains locally on its own traffic-data partition and shares a model update—not raw traffic records—with the server.
- The server combines updates from selected clients using FedAvg to improve the shared forecasting model.

Communication-Aware Selection

- FlowPulse-FL compares full participation, random client selection, and FlowPulse client selection.
- In the evaluated setup, selecting 2 of 4 clients per round reduces communication from 19,200 bytes to 9,600 bytes across 12 federated rounds—a 50% reduction compared with full participation.

VERIFIED THREE-SEED RESULTS

FlowPulse selection uses 9,600 communication bytes instead of 19,200 bytes with full participation—a 50% reduction.

Accuracy remains comparable across IID and non-IID settings, evaluated over seeds 42, 7, and 2026.



Results | Communication | Convergence

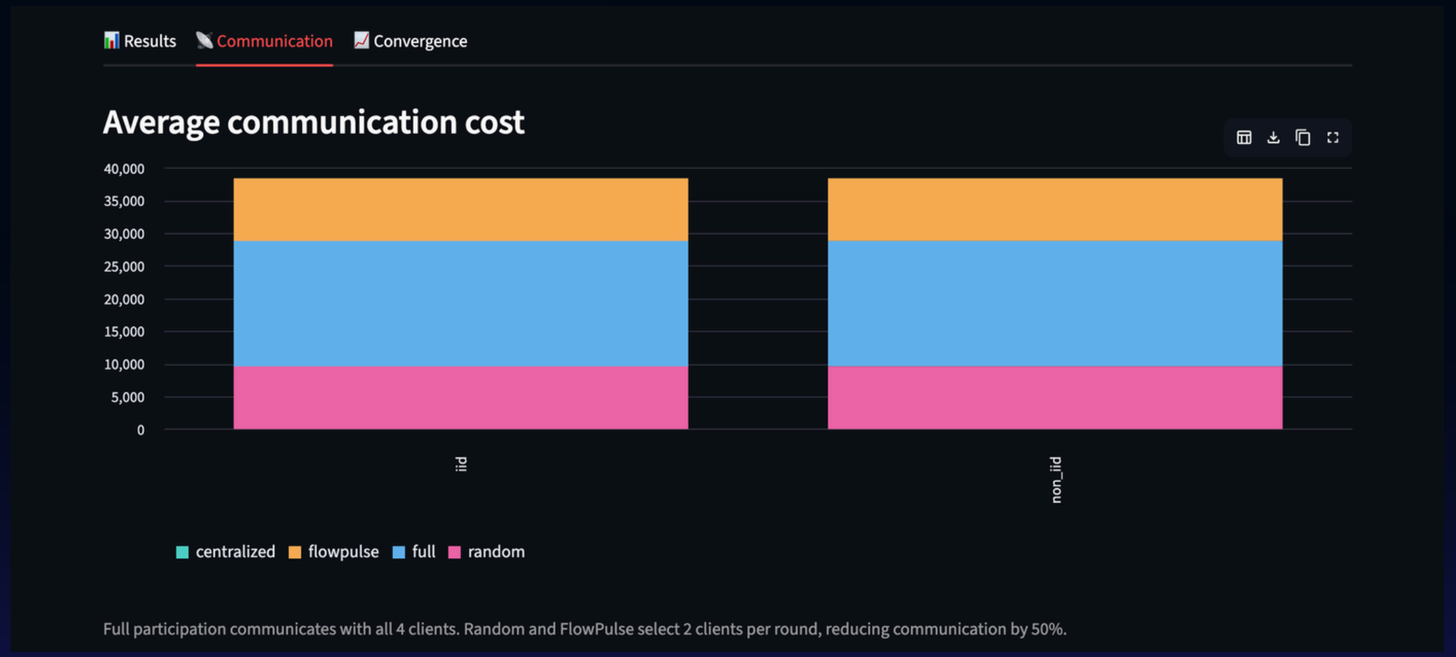
Three-seed forecasting performance

partition_mode	selection_strategy	mae_mean	mae_std	rmse_mean	rmse_std	communication_bytes_mean	communication_bytes_std	runs
iid	centralized	759.1505	10.5023	982.4509	14.7242	0	0	3
iid	flowpulse	759.346	10.2767	982.6267	14.6129	9600	0	3
iid	full	759.1539	10.5006	982.4579	14.7241	19200	0	3
iid	random	758.9825	10.6157	982.4416	14.8412	9600	0	3
non_iid	centralized	759.1505	10.5023	982.4509	14.7242	0	0	3
non_iid	flowpulse	762.66	10.4916	985.2234	13.6228	9600	0	3
non_iid	full	759.5167	10.5128	982.931	14.7148	19200	0	3
non_iid	random	756.3515	8.1718	981.0421	13.7077	9600	0	3

Mean and standard deviation across seeds 42, 7, and 2026.

COMMUNICATION REDUCTION

Selecting 2 of 4 clients per round reduces communication by 50%.
Full participation: 19,200 bytes
FlowPulse selection: 9,600 bytes



CONCLUSION & THANK YOU YOU

FlowPulse-FL demonstrates a communication-aware approach to federated traffic-flow forecasting.

- Selecting 2 of 4 simulated traffic-intersection clients per round reduces communication from 19,200 bytes to 9,600 bytes over 12 federated rounds—a 50% reduction compared with full participation.
- Across seeds 42, 7, and 2026, FlowPulse selection maintains accuracy comparable to full participation and random selection in IID and non-IID settings.

Thank you for reviewing FlowPulse-FL.